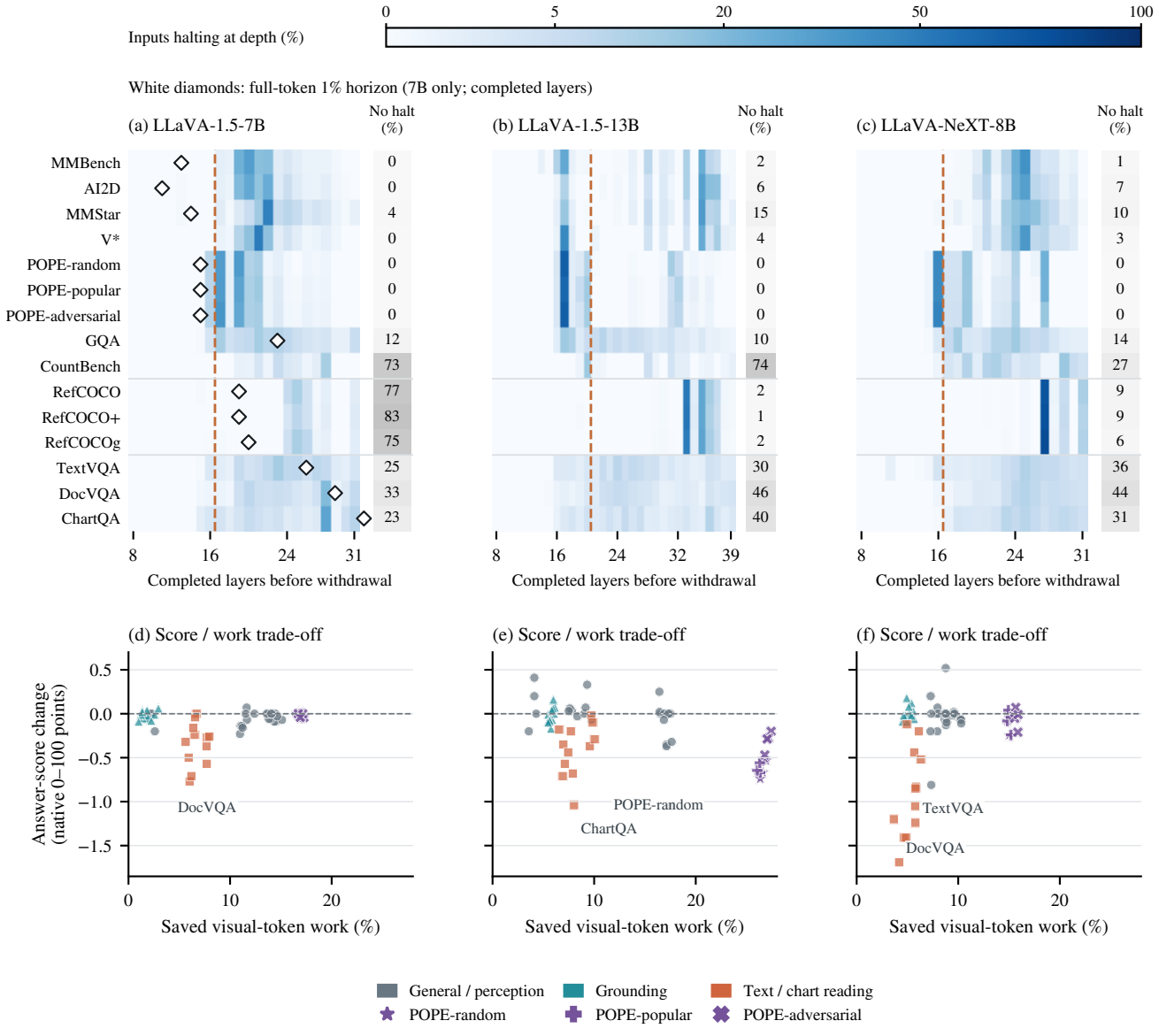
Inputs halting at depth (%)
0
5
20
50
100
White diamonds: full-token 1% horizon (7B only; completed layers)
(a) LLaVA-1.5-7B
(b) LLaVA-1.5-13B
(c) LLaVA-NeXT-8B
No halt (%)
MMBench 0 2 1
AI2D 0 6 7
MMStar 4 15 10
V* 0 4 3
POPE-random 0 0 0
POPE-popular 0 0 0
POPE-adversarial 0 0 0
GQA 12 10 14
CountBench 73 74 27
RefCOCO 77 2 9
RefCOCO+ 83 1 9
RefCOCOg 75 2 6
TextVQA 25 30 36
DocVQA 33 46 44
ChartQA 23 40 31
8 16 24 31
8 16 24 32 39
8 16 24 31
Completed layers before withdrawal
(d) Score / work trade-off
(e) Score / work trade-off
(f) Score / work trade-off
Answer-score change (native 0–100 points)
0.5
0.0
−0.5
−1.0
−1.5
DocVQA
POPE-random
ChartQA
TextVQA
DocVQA
0
10
20
Saved visual-token work (%)
General / perception
Grounding
Text / chart reading
POPE-random
POPE-popular
POPE-adversarial